User Material UMAT Implementation of Nearly Incompressible Anisotropic GOH Model in ABAQUS

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Statement

ChatGPT assisted in preparing this document and its MATLAB and UMAT Fortran subroutine. Although reasonable care has been taken to ensure accuracy, readers should independently verify the results and report any issues. This work serves as a transparent reference for teaching, model verification, and UMAT development, rather than a substitute for rigorous validation.

1 Objective

The following section summarizes the formulation and ABAQUS/Standard UMAT implementation of an anisotropic Gasser-Ogden-Holzapfel (GOH) model combined with a tension-compression switch. This example illustrates how to handle material anisotropy in an ABAQUS UMAT.

2 Gasser-Ogden-Holzapfel (GOH) Model Formulation

2.1 Strain Energy Density Function

The GOH model describes the mechanical response of anisotropic soft tissues by decomposing the strain energy density into an isotropic matrix contribution, an anisotropic fiber contribution, and a volumetric contribution:

$$\Psi = c_{10}(\bar{I}_1 - 3) + \sum_{\alpha} \frac{k_{1\alpha}}{2k_{2\alpha}} \left[\exp\left(k_{2\alpha}\langle E_\alpha \rangle^2\right) - 1\right] + \frac{1}{D}\left(\frac{J^2-1}{2} - \ln J\right). \tag{1}$$

Here, c_{10} is the isotropic matrix stiffness parameter, $k_{1\alpha}$ and $k_{2\alpha}$ describe the fiber stiffness and nonlinear strain-stiffening behavior, respectively, and D is the compressibility parameter. The fiber strain measure E_α is defined as

$$E_\alpha = \kappa_\alpha(\bar{I}_1 - 3) + (1 - 3\kappa_\alpha)(\bar{I}_{4\alpha} - 1), \tag{2}$$

where κ_α represents the fiber dispersion parameter. The modified invariants are defined as

$$\bar{I}_1 = J^{-2/3} I_1, \tag{3}$$

$$\bar{I}_{4\alpha} = J^{-2/3} I_{4\alpha}, \tag{4}$$

where

$$I_1 = \mathrm{tr}(\mathbf{C}), \tag{5}$$

and

$$I_{4\alpha} = \mathbf{N}_\alpha \cdot \mathbf{C}\mathbf{N}_\alpha. \tag{6}$$

Here, $\mathbf{N}_\alpha$ denotes the reference fiber direction vector and $\mathbf{C}$ is the right Cauchy–Green deformation tensor. For convenience, the structural tensor associated with fiber family α is defined as

$$\mathbf{M}_\alpha = \mathbf{N}_\alpha \otimes \mathbf{N}_\alpha, \tag{7}$$

such that

$$I_{4\alpha} = \mathbf{M}_\alpha : \mathbf{C}. \tag{8}$$

The Macaulay bracket is defined by

$$\langle E_\alpha \rangle = \max(E_\alpha, 0), \tag{9}$$

which ensures that the fiber contribution is activated only when the fibers are stretched. The corresponding Heaviside function is defined as

$$H(E_\alpha) = \begin{cases} 1, & E_\alpha > 0, \\ 0, & E_\alpha < 0. \end{cases} \tag{10}$$

It should be noted that this variable is used to determine whether the fiber is activated. However, it is neither an internal variable nor a history-dependent quantity. The activation state of the fiber depends only on the current deformation and is independent of its previous activation history. Therefore, no state variable is required in the implementation. This tension–compression switch is essential for the GOH model, as it ensures that the fiber contribution is activated only under tension. However, the direct switching between the active and inactive states results in a discontinuous change in the tangent stiffness, which may cause numerical difficulties during the finite element analysis.

2.2 Auxiliary Derivatives

The deformation gradient determinant J is related to the right Cauchy–Green tensor by

$$J = \sqrt{\det(\mathbf{C})}. \tag{11}$$

The first derivative of J with respect to the symmetric tensor $\mathbf{C}$ is

$$\frac{\partial J}{\partial \mathbf{C}} = \frac{J}{2}\mathbf{C}^{-1}. \tag{12}$$

The derivative of the inverse tensor with respect to the symmetric tensor $\mathbf{C}$ is

$$\frac{\partial \mathbf{C}^{-1}}{\partial \mathbf{C}} = -\mathbf{C}^{-1} \odot \mathbf{C}^{-1}. \tag{13}$$

Consequently, the second derivative of J is

$$\frac{\partial^2 J}{\partial \mathbf{C} \partial \mathbf{C}} = \frac{J}{4}\mathbf{C}^{-1} \otimes \mathbf{C}^{-1} - \frac{J}{2}\mathbf{C}^{-1} \odot \mathbf{C}^{-1}. \tag{14}$$

where

$$\left(\mathbf{C}^{-1} \otimes \mathbf{C}^{-1}\right)_{ijkl} = C^{-1}_{ij} C^{-1}_{kl}$$

$$\left(\mathbf{C}^{-1} \odot \mathbf{C}^{-1}\right)_{ijkl} = \frac{1}{2}\left(C^{-1}_{ik} C^{-1}_{jl} + C^{-1}_{il} C^{-1}_{jk}\right)$$

2.2.1 Derivative of the Modified First Invariant

The first derivative of the modified first invariant is

$$\frac{\partial \bar{I}_1}{\partial \mathbf{C}} = J^{-2/3}\left(\mathbf{I} - \frac{1}{3} I_1 \mathbf{C}^{-1}\right). \tag{15}$$

The second derivative of the modified first invariant is

$$\frac{\partial^2 \bar{I}_1}{\partial \mathbf{C}\partial \mathbf{C}} = J^{-2/3}\left[-\frac{1}{3}\left(\mathbf{C}^{-1} \otimes \mathbf{I} + \mathbf{I} \otimes \mathbf{C}^{-1}\right) + \frac{1}{9} I_1 \mathbf{C}^{-1} \otimes \mathbf{C}^{-1} + \frac{1}{3} I_1 \mathbf{C}^{-1} \odot \mathbf{C}^{-1}\right]. \tag{16}$$

2.2.2 Derivative of the Modified Fiber Invariant

The first derivative of the modified fiber invariant is

$$\frac{\partial \bar{I}_{4\alpha}}{\partial \mathbf{C}} = J^{-2/3}\left(\mathbf{M}_\alpha - \frac{1}{3} I_{4\alpha} \mathbf{C}^{-1}\right). \tag{17}$$

The second derivative is

$$\frac{\partial^2 \bar{I}_{4\alpha}}{\partial \mathbf{C}\partial \mathbf{C}} = J^{-2/3}\left[-\frac{1}{3}\left(\mathbf{C}^{-1} \otimes \mathbf{M}_\alpha + \mathbf{M}_\alpha \otimes \mathbf{C}^{-1}\right) + \frac{1}{9} I_{4\alpha} \mathbf{C}^{-1} \otimes \mathbf{C}^{-1} + \frac{1}{3} I_{4\alpha} \mathbf{C}^{-1} \odot \mathbf{C}^{-1}\right]. \tag{18}$$

2.2.3 Derivative of the Fiber Strain Measure

The first derivative of E_α is

$$\frac{\partial E_\alpha}{\partial \mathbf{C}} = \kappa_\alpha \frac{\partial \bar{I}_1}{\partial \mathbf{C}} + (1 - 3\kappa_\alpha)\frac{\partial \bar{I}_{4\alpha}}{\partial \mathbf{C}}. \tag{19}$$

The second derivative of E_α is

$$\frac{\partial^2 E_\alpha}{\partial \mathbf{C}\partial \mathbf{C}} = \kappa_\alpha \frac{\partial^2 \bar{I}_1}{\partial \mathbf{C}\partial \mathbf{C}} + (1 - 3\kappa_\alpha)\frac{\partial^2 \bar{I}_{4\alpha}}{\partial \mathbf{C}\partial \mathbf{C}}. \tag{20}$$

Substituting the preceding expressions gives

$$\begin{aligned} \frac{\partial^2 E_\alpha}{\partial \mathbf{C}\partial \mathbf{C}} = J^{-2/3}\Bigg[& -\frac{\kappa_\alpha}{3}\left(\mathbf{C}^{-1} \otimes \mathbf{I} + \mathbf{I} \otimes \mathbf{C}^{-1}\right) \\ & - \frac{1 - 3\kappa_\alpha}{3}\left(\mathbf{C}^{-1} \otimes \mathbf{M}_\alpha + \mathbf{M}_\alpha \otimes \mathbf{C}^{-1}\right) \\ & + \frac{1}{9}\left[\kappa_\alpha I_1 + (1 - 3\kappa_\alpha) I_{4\alpha}\right] \mathbf{C}^{-1} \otimes \mathbf{C}^{-1} \\ & + \frac{1}{3}\left[\kappa_\alpha I_1 + (1 - 3\kappa_\alpha) I_{4\alpha}\right] \mathbf{C}^{-1} \odot \mathbf{C}^{-1}\Bigg]. \end{aligned} \tag{21}$$

2.3 Second Piola–Kirchhoff Stress

The second Piola–Kirchhoff stress tensor is obtained from the strain energy function according to

$$\mathbf{S} = 2\frac{\partial \Psi}{\partial \mathbf{C}} = \mathbf{S}_{\text{iso}} + \mathbf{S}_{\text{aniso}} + \mathbf{S}_{\text{vol}} \tag{22}$$

2.3.1 Isotropic Matrix Stress

The isotropic matrix stress contribution is

$$\mathbf{S}_{\text{iso}} = 2c_{10}\frac{\partial \bar{I}_1}{\partial \mathbf{C}} \tag{23}$$

Therefore,

$$\mathbf{S}_{\text{iso}} = 2c_{10}J^{-2/3}\left(\mathbf{I} - \frac{1}{3}I_1\mathbf{C}^{-1}\right) \tag{24}$$

2.3.2 Anisotropic Fiber Stress

For $E_\alpha > 0$, the derivative of the fiber energy with respect to E_α is

$$\frac{\partial \Psi_{\text{fiber},\alpha}}{\partial E_\alpha} = k_{1\alpha}E_\alpha \exp(k_{2\alpha}E_\alpha^2). \tag{25}$$

Therefore, the anisotropic fiber stress contribution is

$$\mathbf{S}_{\text{aniso}} = \sum_\alpha H(E_\alpha)2k_{1\alpha}E_\alpha \exp(k_{2\alpha}E_\alpha^2)\frac{\partial E_\alpha}{\partial \mathbf{C}}. \tag{26}$$

2.3.3 Volumetric Stress

The volumetric stress contribution is

$$\mathbf{S}_{\text{vol}} = \frac{2}{D}(J - J^{-1})\frac{\partial J}{\partial \mathbf{C}}. \tag{27}$$

Using

$$\frac{\partial J}{\partial \mathbf{C}} = \frac{J}{2}\mathbf{C}^{-1}, \tag{28}$$

the volumetric stress becomes

$$\mathbf{S}_{\text{vol}} = \frac{1}{D}(J^2 - 1)\mathbf{C}^{-1}. \tag{29}$$

2.4 Material Tangent Modulus

The material tangent modulus associated with the second Piola–Kirchhoff stress is defined as

$$\mathbb{C} = 2\frac{\partial \mathbf{S}}{\partial \mathbf{C}} = \mathbb{C}_{\text{iso}} + \mathbb{C}_{\text{aniso}} + \mathbb{C}_{\text{vol}}. \tag{30}$$

2.4.1 Isotropic Matrix Tangent Modulus

The isotropic matrix tangent modulus is

$$\mathbb{C}_{\text{iso}} = 4c_{10}\frac{\partial^2 \bar{I}_1}{\partial \mathbf{C}\partial \mathbf{C}}. \tag{31}$$

Substituting the second derivative of $\bar{I}_1$ gives

$$\mathbb{C}_{\text{iso}} = 4c_{10}J^{-2/3}\left[-\frac{1}{3}\left(\mathbf{C}^{-1}\otimes\mathbf{I} + \mathbf{I}\otimes\mathbf{C}^{-1}\right) + \frac{1}{9}I_1\mathbf{C}^{-1}\otimes\mathbf{C}^{-1} + \frac{1}{3}I_1\mathbf{C}^{-1}\odot\mathbf{C}^{-1}\right]. \tag{32}$$

2.4.2 Anisotropic Fiber Tangent Modulus

For $E_\alpha > 0$, the second derivative of the fiber energy with respect to E_α is

$$\frac{\partial^2 \Psi_{\text{fiber},\alpha}}{\partial E_\alpha^2} = k_{1\alpha} \exp(k_{2\alpha} E_\alpha^2) \left(1 + 2k_{2\alpha} E_\alpha^2\right). \tag{33}$$

Consequently, the anisotropic tangent modulus is

$$\begin{aligned} \mathbb{C}_{\text{aniso}} = \sum_\alpha H(E_\alpha) \Bigg[& 4k_{1\alpha} \exp(k_{2\alpha} E_\alpha^2) \left(1 + 2k_{2\alpha} E_\alpha^2\right) \left(\frac{\partial E_\alpha}{\partial \mathbf{C}} \otimes \frac{\partial E_\alpha}{\partial \mathbf{C}} \right) \\ & + 4k_{1\alpha} E_\alpha \exp(k_{2\alpha} E_\alpha^2) \frac{\partial^2 E_\alpha}{\partial \mathbf{C} \partial \mathbf{C}} \Bigg]. \end{aligned} \tag{34}$$

2.4.3 Volumetric Tangent Modulus

The volumetric tangent modulus is obtained from

$$\mathbb{C}_{\text{vol}} = 2 \frac{\partial \mathbf{S}_{\text{vol}}}{\partial \mathbf{C}}. \tag{35}$$

Using

$$\mathbf{S}_{\text{vol}} = \frac{2}{D} (J - J^{-1}) \frac{\partial J}{\partial \mathbf{C}}, \tag{36}$$

the volumetric tangent modulus becomes

$$\mathbb{C}_{\text{vol}} = \frac{4}{D} \left[(1 + J^{-2}) \frac{\partial J}{\partial \mathbf{C}} \otimes \frac{\partial J}{\partial \mathbf{C}} + (J - J^{-1}) \frac{\partial^2 J}{\partial \mathbf{C} \partial \mathbf{C}} \right]. \tag{37}$$

Substituting the derivatives of J gives

$$\begin{aligned} \mathbb{C}_{\text{vol}} = \frac{4}{D} \Bigg[& \frac{J^2}{4} (1 + J^{-2}) \mathbf{C}^{-1} \otimes \mathbf{C}^{-1} \\ & + (J - J^{-1}) \left(\frac{J}{4} \mathbf{C}^{-1} \otimes \mathbf{C}^{-1} - \frac{J}{2} \mathbf{C}^{-1} \odot \mathbf{C}^{-1} \right) \Bigg]. \end{aligned} \tag{38}$$

This expression can be further simplified as

$$\mathbb{C}_{\text{vol}} = \frac{1}{D} \left[\left(2J^2 - 1 + J^{-2}\right) \mathbf{C}^{-1} \otimes \mathbf{C}^{-1} - 2(J^2 - 1) \mathbf{C}^{-1} \odot \mathbf{C}^{-1} \right]. \tag{39}$$

2.5 Complete Stress and Material Tangent

The complete second Piola–Kirchhoff stress tensor is

$$\mathbf{S} = 2c_{10} \frac{\partial \bar{I}_1}{\partial \mathbf{C}} + \sum_\alpha H(E_\alpha) 2k_{1\alpha} E_\alpha \exp(k_{2\alpha} E_\alpha^2) \frac{\partial E_\alpha}{\partial \mathbf{C}} + \frac{1}{D} (J^2 - 1) \mathbf{C}^{-1}. \tag{40}$$

The complete material tangent modulus is

$$\mathbb{C} = \mathbb{C}_{\text{iso}} + \mathbb{C}_{\text{aniso}} + \mathbb{C}_{\text{vol}}. \tag{41}$$

Explicitly,

$$
\begin{aligned}
\mathbb{C} =& 4c_{10}\frac{\partial^2 \bar{I}_1}{\partial \mathbf{C}\partial \mathbf{C}} \\
&+ \sum_{\alpha} H(E_\alpha)\Bigg[4k_{1\alpha}\exp(k_{2\alpha}E_\alpha^2)\left(1+2k_{2\alpha}E_\alpha^2\right)\frac{\partial E_\alpha}{\partial \mathbf{C}} \otimes \frac{\partial E_\alpha}{\partial \mathbf{C}} \\
&\quad + 4k_{1\alpha}E_\alpha \exp(k_{2\alpha}E_\alpha^2)\frac{\partial^2 E_\alpha}{\partial \mathbf{C}\partial \mathbf{C}}\Bigg] \\
&+ \frac{4}{D}\left[(1+J^{-2})\frac{\partial J}{\partial \mathbf{C}} \otimes \frac{\partial J}{\partial \mathbf{C}} + (J-J^{-1})\frac{\partial^2 J}{\partial \mathbf{C}\partial \mathbf{C}}\right].
\end{aligned}
\tag{42}
$$

2.6 Summary of the Required Derivatives

For implementation and verification, the required derivatives can be summarized as

$$\frac{\partial J}{\partial \mathbf{C}} = \frac{J}{2}\mathbf{C}^{-1} \tag{43}$$

$$\frac{\partial^2 J}{\partial \mathbf{C}\partial \mathbf{C}} = \frac{J}{4}\mathbf{C}^{-1} \otimes \mathbf{C}^{-1} - \frac{J}{2}\mathbf{C}^{-1} \odot \mathbf{C}^{-1} \tag{44}$$

$$\frac{\partial \bar{I}_1}{\partial \mathbf{C}} = J^{-2/3}\left(\mathbf{I} - \frac{1}{3}I_1\mathbf{C}^{-1}\right) \tag{45}$$

$$\frac{\partial^2 \bar{I}_1}{\partial \mathbf{C}\partial \mathbf{C}} = J^{-2/3}\left[-\frac{1}{3}\left(\mathbf{C}^{-1} \otimes \mathbf{I} + \mathbf{I} \otimes \mathbf{C}^{-1}\right) + \frac{1}{9}I_1\mathbf{C}^{-1} \otimes \mathbf{C}^{-1} + \frac{1}{3}I_1\mathbf{C}^{-1} \odot \mathbf{C}^{-1}\right] \tag{46}$$

$$\frac{\partial \bar{I}_{4\alpha}}{\partial \mathbf{C}} = J^{-2/3}\left(\mathbf{M}_\alpha - \frac{1}{3}I_{4\alpha}\mathbf{C}^{-1}\right) \tag{47}$$

$$\frac{\partial^2 \bar{I}_{4\alpha}}{\partial \mathbf{C}\partial \mathbf{C}} = J^{-2/3}\left[-\frac{1}{3}\left(\mathbf{C}^{-1} \otimes \mathbf{M}_\alpha + \mathbf{M}_\alpha \otimes \mathbf{C}^{-1}\right) + \frac{1}{9}I_{4\alpha}\mathbf{C}^{-1} \otimes \mathbf{C}^{-1} + \frac{1}{3}I_{4\alpha}\mathbf{C}^{-1} \odot \mathbf{C}^{-1}\right] \tag{48}$$

$$\frac{\partial E_\alpha}{\partial \mathbf{C}} = \kappa_\alpha \frac{\partial \bar{I}_1}{\partial \mathbf{C}} + (1-3\kappa_\alpha)\frac{\partial \bar{I}_{4\alpha}}{\partial \mathbf{C}} \tag{49}$$

$$\frac{\partial^2 E_\alpha}{\partial \mathbf{C}\partial \mathbf{C}} = \kappa_\alpha \frac{\partial^2 \bar{I}_1}{\partial \mathbf{C}\partial \mathbf{C}} + (1-3\kappa_\alpha)\frac{\partial^2 \bar{I}_{4\alpha}}{\partial \mathbf{C}\partial \mathbf{C}} \tag{50}$$

All derivatives above are defined on the space of symmetric second-order tensors, and therefore the symmetry of $\mathbf{C}$ is explicitly incorporated through $\mathbb{I}^{\mathrm{s}}$ and $\odot$.

3 Code Implementation

ChatGPT was used to assist with the implementation of the GOH model. With a concise description of the required constitutive formulation and coding style, a complete UMAT implementation can be generated. The resulting code is then independently checked and validated through an ABAQUS/Standard simulation and comparison with a semi-analytical solution.

- **Fiber orientation definition:**

  The fiber orientation defines the preferred material directions associated with the anisotropic response of the GOH model. In the present implementation, the fiber directions are directly defined within the UMAT rather than through the ABAQUS input file. This approach is appropriate when the fiber orientation is spatially uniform and fixed in the reference configuration. Thus, the same fiber directions are assigned to all integration points. The fiber directions are specified by the unit vectors $\mathbf{N}_\alpha$ in the reference configuration, where $\alpha = 1, 2$ denotes the two fiber families. That is

$$\mathbf{N}_1 = \begin{bmatrix} \cos\beta \\ \sin\beta \\ 0 \end{bmatrix}, \qquad \mathbf{N}_2 = \begin{bmatrix} \cos\beta \\ -\sin\beta \\ 0 \end{bmatrix}, \tag{51}$$

  where β is the angle between the fiber direction and the global x-axis.

- **Structural tensor:**

  The corresponding structural tensor is defined as

$$\mathbf{M}_\alpha = \mathbf{N}_\alpha \otimes \mathbf{N}_\alpha, \qquad \alpha = 1, 2. \tag{52}$$

  The structural tensor characterizes the preferred material direction and enters the anisotropic response through the fiber invariant

$$I_{4\alpha} = \mathbf{N}_\alpha \cdot \mathbf{C}\mathbf{N}_\alpha = \mathbf{M}_\alpha : \mathbf{C}. \tag{53}$$

- **Physical meaning of the fiber direction:**

  The fiber direction represents the preferred material direction at each material point rather than a single physical fiber extending throughout the entire body. In other words, it characterizes the directional reinforcement of the material at each point. When the same fiber orientation is assigned to all material points, the material remains homogeneous but is anisotropic.

- **When should ABAQUS *orientation be used?**

  The direct definition of $\mathbf{N}_\alpha$ inside the UMAT is appropriate when the preferred material directions are prescribed and spatially uniform. An ABAQUS material ***orientation** should instead be considered when the material orientation varies spatially, when different regions of the model have different material directions, or when the constitutive directions need to be defined with respect to a local coordinate system.

- **Homogeneous deformation:**

  Although the material is anisotropic, a spatially uniform fiber orientation does not necessarily lead to an inhomogeneous deformation. Under simple deformation modes and appropriate homogeneous boundary conditions, the deformation can remain homogeneous. Therefore, the presence of fiber families does not by itself prevent the existence of a homogeneous deformation.

- **Analytical verification:**

  When the deformation is homogeneous, the constitutive response can be evaluated without solving a spatially varying boundary-value problem. Therefore, the ABAQUS/Standard UMAT implementation can be verified against an analytical or semi-analytical solution.

- **Semi-analytical solution for the compressible GOH model:**

  For the compressible GOH model considered here, a fully explicit analytical solution for the uniaxial stress–stretch relation is difficult to obtain because the lateral stretches are coupled with the volumetric and anisotropic contributions. Nevertheless, a semi-analytical solution can be obtained by prescribing the axial stretch and solving the lateral traction-free conditions numerically, while evaluating the constitutive equations analytically. This solution provides an independent reference for validating the UMAT implementation.

4 GOH UMAT Validation

4.1 Validation Technique

The anisotropic characteristics of the GOH model are verified by changing the fiber orientation while keeping all other material parameters unchanged. For two symmetric fiber families, the angle β can be varied over a representative range, for example,

$$\beta = 0^\circ, \quad 15^\circ, \quad 30^\circ, \quad 60^\circ, \quad 90^\circ. \tag{54}$$

The resulting stress–stretch curves are compared to confirm that the material response changes consistently with the fiber orientation. This test verifies that the structural tensors $\mathbf{M}_\alpha$ and the corresponding fiber invariants $I_{4\alpha}$ are correctly implemented. See Fig.1.

The symmetry of the two fiber families is verified independently. For the two families oriented at $+\beta$ and $-\beta$, the material response should satisfy the corresponding symmetry requirements. In particular, under uniaxial tension along the x-direction, the symmetric fiber arrangement should result in

$$\sigma_{12} = 0. \tag{55}$$

Furthermore, changing the fiber orientation from $+\beta$ to $-\beta$ should not change the axial stress–stretch response. This test provides a direct verification of the implementation of the two fiber families and their structural tensors.

4.2 The Results Comparison

The ABAQUS/Standard results are compared with the corresponding semi-analytical solution for the homogeneous uniaxial tensile deformation. The specimen has an initial length of $L_0 = 10$. The uniaxial stretch is varied from $\lambda = 1$ to $\lambda = 2$. The loading direction is the global x-direction. Two symmetric fiber families are considered with β. The agreement between the two results provides a direct verification of the stress calculation and the constitutive implementation in the UMAT.

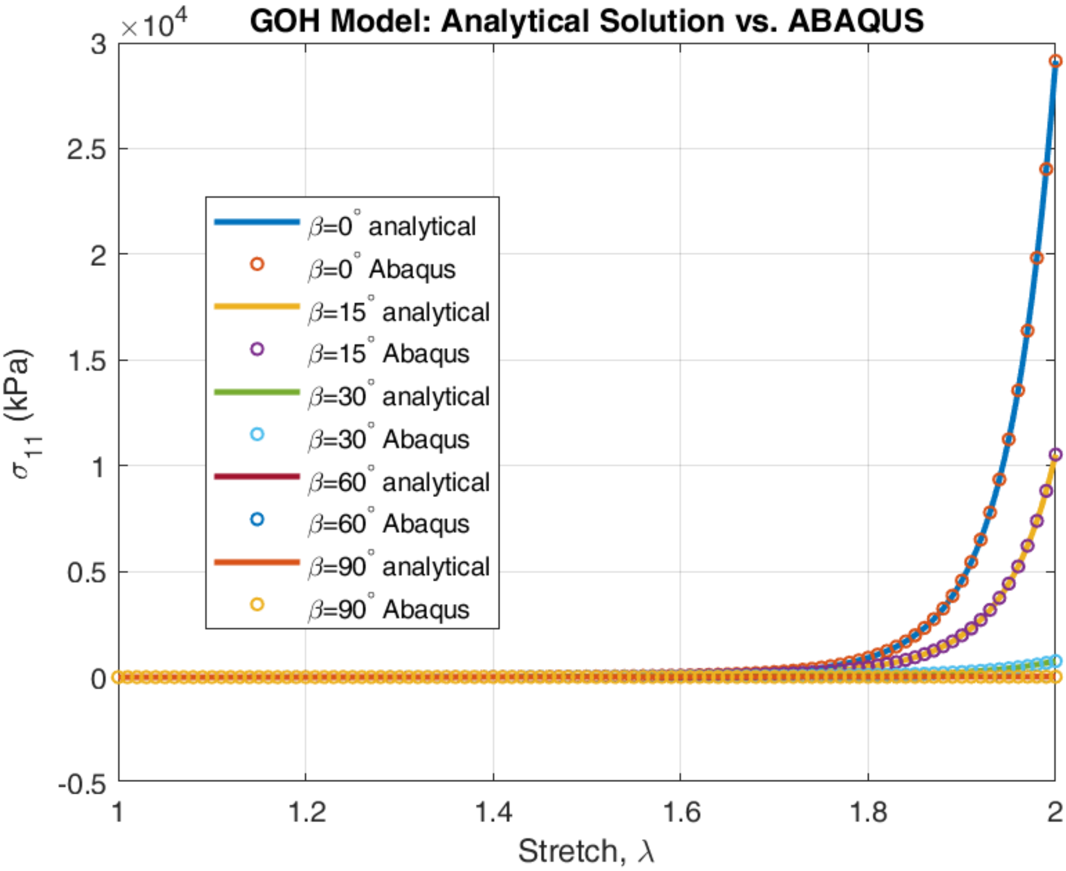

Figure 1: Comparison between the ABAQUS/Standard UMAT results and the semi-analytical solution of the GOH model under uniaxial tension.

5 Appendix

The detailed UMAT code implementation, simple uniaxial tension example in ABAQUS, and analytical solution in Matlab can be found on GitHub: https://github.com/Xuefengyuki/UMAT_NearlyIncompressibleYeohFlemingChagnonDamageMullinsEffect.git